

Load Interconnection System Impact Study Report

For Queue Project Q22-2060

Wilsonville 230 kV

August 21, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2060. The Interconnection Customer proposes a 682.5 MW large load (data center) facility interconnecting at the Wilsonville 230 kV substation in the Indiana area. The request was received on April 08, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 617 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$55,895,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Eastown Road 100 kV/1 kV TX	George Neal South Substation to Myrtlept Substation 230 KV	139.3	267	7.2
Stmarys to Leyendecker 100 KV	East Drainage Canal to St. Lawrence 230 KV	132.7	240	31.8
St. Lawrence to E Drainage Canal 230 KV	James De Young to St. Lawrence 230 KV	125.8	520	54.5
Eastown Rd to Geisel Monument 100 KV	Myrtle Pt to George Neal South 230 KV	120.1	126	8.1
East Drainage Canal to St Marys 100 KV Cir. 2	E Drainage Canal to St Marys 100 KV Ckt 1	116.0	246	17.0
St Marys Substation to East Drainage Canal 100 KV Circuit 1	E Drainage Canal to St Marys Substation 100 KV #2	113.7	251	17.0
Pruitt Escalante Pegs to Taconite Harbor Energy Center 100 KV L	E Drainage Canal Substation to St. Lawrence 230 KV	111.1	131	13.0
Eastown Rd 230/1 kV XFMR	George Neal S to Myrtle Point 230 KV	108.3	366	7.2
Clarence to Pruitt Escalante Pegs 100 KV	St. Lawrence to East Drainage Canal 230 KV	106.3	123	13.0
Ephraim Cemetery Substation to Clarence 100 KV	St. Lawrence Substation to E Drainage Canal 230 KV	102.4	115	13.0
Ephraim Cemetery to Taiban Mesa 100 KV	E Drainage Canal to St. Lawrence 230 KV line	102.2	109	13.0
Dutton Pump to Bourne 100 KV	East Drainage Canal to St. Lawrence 230 KV Line	100.9	229	13.8
Klamath Co-Gen to Bourne 100 KV Line	E Drainage Canal to St. Lawrence 230 KV	100.8	244	13.8
Klamath Co-Gen to Sherman St 100 KV	St. Lawrence Substation to East Drainage Canal 230 KV	100.6	255	13.8

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Sherman St to Eastmont 100 KV line	E Drainage Canal to St. Lawrence 230 KV	100.5	265	13.8
Pomaria to Taiban Mesa 100 KV	E Drainage Canal to St. Lawrence 230 KV line	100.4	163	20.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Easttown Road 100 kV/1 kV TX	Replace transformer with higher-rated unit	\$4,693,000	26% (\$1,236,000)
Stmarys to Leyendecker 100 KV	Rebuild line with higher-capacity conductor	\$645,000	71% (\$458,000)
E Drainage Canal to St. Lawrence 230 KV	Rebuild line with higher-capacity conductor	\$34,374,000	86% (\$29,556,000)
Easttown Road to Geisel Monument 100 KV Line	Rebuild line with higher-capacity conductor	\$1,895,000	38% (\$721,000)
St Marys to East Drainage Canal 100 KV #2	Rebuild line with higher-capacity conductor	\$571,000	56% (\$319,000)
East Drainage Canal to St Marys 100 KV #1	Rebuild line with higher-capacity conductor	\$750,000	56% (\$419,000)
Pruitt Escalante Pegs to Taconite Harbor Energy Center 100 KV L	Rebuild line with higher-capacity conductor	\$7,742,000	94% (\$7,246,000)
Easttown Road 230/1kV Transformer	Replace transformer with higher-rated unit	\$4,879,000	26% (\$1,285,000)
Clarence to Pruitt Escalante Pegs 100 KV	Reconductor line	\$2,214,000	94% (\$2,072,000)
Ephraim Cemetery Substation to Clarence 100 KV	Reconductor line	\$4,764,000	94% (\$4,459,000)
Ephraim Cemetery to Taiban Mesa 100 KV	Reconductor line	\$1,963,000	94% (\$1,837,000)
Dutton Pump to Bourne 100 KV	Reconductor line	\$807,000	46% (\$372,000)
Klamath Co-Gen to Bourne 100 KV Line	Reconductor line	\$2,649,000	46% (\$1,222,000)
Klamath Co-Gen to Sherman St 100 KV	Reconductor line	\$2,818,000	46% (\$1,301,000)
Sherman St to Eastmont 100 KV Line	Reconductor line	\$1,406,000	46% (\$649,000)
Pomaria to Taiban Mesa 100 KV	Reconductor line	\$3,086,000	89% (\$2,743,000)
	Total allocated to project		\$55,895,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.